

LUT core-module ablation: three generations of lookup math

Comparison table

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Protocol

Architecture: LUT transformer, $E = 384$, 6 layers, 6 attention heads. FFN = `CompressionMultiHeadLUT` with $H = 4$ heads $\times$ $tph = 128$ tables ($P = H \cdot tph = 512$ tables per token), $d_{in} = d_{out} = 48$, $n = 8$ anchor pairs per table ($K = 2^n = 256$ cells); `compress` $384 \rightarrow 192$, `decompress` $192 \rightarrow 384$. **Training:** 16,000 steps, batch size 48 (24576 tokens), untied embeddings, LUT tables without weight decay. **Eval:** `evaluate_bpb_fixed`, 48 rows $\times$ 100 steps, first 12 rows skipped.

Counting convention

- Core LUT only, per token, one layer. The `compress/decompress` projections are identical in every row and excluded (147,456 MACs forward).
- **FLOP** = one scalar float operation ($+$, $-$, $\times$, $\div$, a comparison, $|\cdot|$, $\exp$, $\log \sigma$, σ). **MAC** = a multiply accumulated into a sum; 1 MAC = 2 FLOPs, included in the FLOP column. Memory reads and integer work are not counted.
- **Forward** = inference. **Backward** = all additional training arithmetic, including per-table gradients of per-layer scalars.

Table A — forward function and gradient function

Per variant, (a) is the function computed on the forward pass and (b) the function whose gradient training uses: the weights always receive the gradient of (a), while the input, the temperatures and the learned per-layer scalars receive the gradient of (b) with the table weights held constant.

Variant	(a) Forward: the function computed	(b) Backward: the function whose gradient is used
GEN 1 — Spiking Manifesto basic model		
1.1 Hard forward, two-alternatives backward	$y = \sum_t W_t[c_t]$, summed over tables t : each of table t 's n anchor pairs gives a signed margin $u_j = z[a_j] - z[b_j]$; the n signs $[u_j > 0]$ are the bits of the binary index $c_t \in \{0, \dots, 2^n - 1\}$ of the row read from $W_t \in \mathbb{R}^{K \times d}$, $K = 2^n$.	The 2-cell blend $\sum_t [(1 - U_t)W_t[c_t] + U_tW_t[c'_t]]$, $U_t = U(u_{j^*})$, $U(u) = 0.5/(1 + u)$, with the least-confident pair $j^* = \arg \min_j u_j $ and c' (c with bit j^* flipped) held fixed. The pairing with (a) is exact as a stop-gradient composite; the blend is C^1 across a bit flip and jumps only where j^* switches.
1.2 Soft forward, two-alternatives backward	(a) = (b): the blend of 1.1(b), applied at eval too.	
GEN 2 — FastMHL math		
2.1 Hard forward, all-alternatives backward (all K cells)	$y = \sum_t W_t[c_t]$, as 1.1(a).	With $b_j(\ell) = +1$ if the bit of ℓ for pair j is 1, else -1 , and the squashed margin $\hat{m}_j = u_j/(T_s + u_j) \in (-1, 1)$: $s_\ell = \sum_j b_j(\ell) \hat{m}_j$ scores how well row ℓ 's bit pattern agrees with the actual signs of the margins (small margins count little either way). All K rows of each table are read, softmax-weighted within that table: $\sum_t \sum_{\ell < K} \text{softmax}_\ell(s_\ell/T_{sel}) W_t[\ell]$. T_s, T_{sel} : per-layer learned scalars. Globally C^1 in the input; never uses the address, whose row is the one with the largest weight.
2.2 Hybrid_smooth two-alternatives forward, all-alternatives backward	Reads two cells, the addressed c_t and the alternative c'_t (c_t with the bit of the least-confident pair $j^* = \arg \min_j u_j $ flipped), blended by a weight set by that pair's margin: $y = \sum_t [(1 - w_t)W_t[c_t] + w_tW_t[c'_t]]$, $w_t = \sigma(-2\rho_{j^*}/T_{sel}) \in (0, \frac{1}{2}]$, $\rho_j = u_j /(T_s + u_j)$, σ the logistic sigmoid; applied at eval too.	The same softmax-weighted read of all K rows as 2.1(b), with the same T_s, T_{sel} parameters as (a).
2.3 Hard forward, 2-alternatives backward	The hard read of 1.1(a).	The softmax of 2.1(b) restricted to the two cells c_t and c'_t (c_t with the bit of the least-confident pair $j^* = \arg \min_j u_j $ flipped) and renormalised over that pair. The weight on c'_t is $\sigma(-2\rho_{j^*}/T_{sel})$, $\rho_j = u_j /(T_s + u_j)$, σ the logistic sigmoid: exactly 2.2(a)'s w_t , so this is 2.2(a)'s 2-cell blend.
2.4 Hybrid_smooth two-alternatives forward, 2-alternatives backward	(a) = (b): the 2-cell blend of 2.2(a). The backward keeps only 2 of the K rows, the addressed c_t and its least-confident neighbour c'_t , which are exactly the two cells the forward reads, so the restricted softmax equals the forward weight w_t and forward and backward coincide; unlike 2.2, the input gradient reaches only pair j^* .	
GEN 3 — LightMHL math (LookupFFN-inspired): learned margin score		
3.1 The Gen-3 form: learned margin score, one cell, per-layer (β, γ)	(a) = (b): $y = \sum_t s_t W_t[c_t]$ with the address c_t held fixed; $s_t = S_t P_t^\gamma$, $S_t = \sum_j m_j$, $P_t = \prod_j \sigma(\beta m_j)$, $m_j = u_j $, computed as $S_t \exp(\gamma \sum_j \log \sigma(\beta m_j))$; β, γ learned per layer from (2, 1), where s_t is LookupFFN's margin score. Discontinuous at cell boundaries, since $\sigma(0) > 0$ (trained jump: median 2.22% of $\ y\ $).	
3.2 The Gen-3 form, two alternatives, per-layer (β, γ, τ)	(a) = (b): each table reads two cells, $y = \sum_t s_t [(1 - v_t)W_t[c_t] + v_tW_t[c'_t]]$, with s_t as in 3.1 over all n margins and c'_t the cell c_t with the bit of the smallest margin $j^* = \arg \min_j m_j$ flipped (c_t, c'_t, j^* held fixed). v_t is a softmax over the two cells with logits 0 and $-2m_{j^*}/\tau$ (flipping bit j lowers $\langle u, b \rangle$ by $2m_j$), i.e. $v_t = \sigma(-2m_{j^*}/\tau) \in (0, \frac{1}{2}]$; τ learned per layer from 0.5. The input gradient reaches all n margins through s_t and m_{j^*} alone through v_t . Continuous across a bit flip ($v_t = \frac{1}{2}$ there); jumps where j^* switches.	

Table B — cost and results

Per token, one layer, core LUT only; $P = 512$, $n = 8$, $K = 256$, $d = 48$.

Variant	Forward FLOPs	Backward FLOPs	Forward MACs	Backward MACs	Validation bpb	Implementation (class used)
1.1	$P(2n + d) = \mathbf{32,768}$	$P(5d + 2n + 9) = \mathbf{135,680}$	0	$P \cdot 2d = \mathbf{49,152}$	TBD — not yet implemented	MultiHeadLut(smooth_mode=False, n_alternatives=1, uncertainty_mode=INVERSE_L1); not wired into CompressionMultiHeadLUT.
1.2	$P(4d + 4n + 4) = \mathbf{116,736}$	$P(8d + 9) = \mathbf{201,216}$	$P \cdot 2d = \mathbf{49,152}$	$P \cdot 4d = \mathbf{98,304}$	TBD — not yet implemented	MultiHeadLut(smooth_mode=True, n_alternatives=1); not wired in.
2.1	$P(2n + d) = \mathbf{32,768}$	$P(2Kd + 4nK + 9K + 11n + d) = \mathbf{18,026,496^e}$	0	$P(Kd + 2nK) = \mathbf{8,388,608}$	1.180622 (exp_n_0121) ^f	FastMultiHeadLut(forward_mode='hard', backward_topk=0).
2.2	$P(2kd + 4n + 5) = \mathbf{117,248}$, $k = 2$ cells read	$P(2Kd + 4nK + 9K + 11n + 2kd) = \mathbf{18,100,224^e}$	$Pkd = \mathbf{49,152}$	$P(Kd + 2nK + kd) = \mathbf{8,437,760}$	TBD — not yet run	FastMultiHeadLut(forward_mode='hybrid_smooth', backward_topk=0).
2.3	$P(2n + d) = \mathbf{32,768}$	$P(2kd + 4nk + 9k + 12n + d - 1) = \mathbf{213,504}$, $k = 2$ cells	0	$P(kd + 2nk) = \mathbf{65,536}$	TBD — not yet run	FastMultiHeadLut(forward_mode='hard', backward_topk=1); backward_topk counts the cells kept besides c_t .
2.4	$P(2kd + 4n + 5) = \mathbf{117,248}$	$P(4kd + 9n + 15) = \mathbf{241,152}$	$Pkd = \mathbf{49,152}$	$P2kd = \mathbf{98,304}$	TBD — not yet run	FastMultiHeadLut(forward_mode='hybrid_smooth', backward_topk=1).
3.1	$P(2kd + 7n + 2) = \mathbf{78,848}$, $k = 1^g$	$P(4kd + 10n + 8) = \mathbf{143,360^g}$	$Pkd = \mathbf{24,576}$	$P2kd = \mathbf{49,152}$	1.169328 (exp_g_0248)	LightMultiHeadLUT(confidence_form='learned_margin', learned_margin_init=(0, 2, 1), learned_margin_freeze_g=True).
3.2	$P(2kd + 8n + 9) = \mathbf{135,680}$, $k = 2^g$	$P(4kd + 10n + 23) = \mathbf{249,344^g}$	$Pkd = \mathbf{49,152}$	$P2kd = \mathbf{98,304}$	TBD — not yet run	As 3.1 with read_top_n=2, read_tau=0.5, read_tau_learnable=True.

Notes to Table B.

^e Dominated by $2Kd$: every table's K rows are read; several [tokens, 512, 256] fp32 buffers of about 12.9 GiB at 24,576 tokens.

^f Re-score of the saved run under the corrected eval (0 missing keys); its in-run batch-coupled number was 1.191455.

^g Per table, forward: n comparisons, n differences, n absolute values, n products βm_j , $n \log \sigma$, the two sums, $\times \gamma$, $+g$ (still executed, $g = 0$), exp and the final product $(7n + 2)$, plus the scaled rows $(2kd)$; 3.2 adds the argmin $(n - 1)$, the logit $-2m_{j^*}/\tau$ (2), the two-way softmax (4) and s_t times the two weights (2). Backward: the row terms $(4kd)$, the input term $(8n + 3)$ and the per-table accumulation of the parameter gradients $(\partial \log \gamma$ 3, $\partial \log \beta$ $2n + 2$); 3.2 adds ∂s_t from two rows (3), ∂v_t (2), the softmax backward (5), ∂m_{j^*} (3) and $\partial \log \tau$ (2). The parameter gradients are scalar multiplies, so they add no MACs.

The Gen-3 form

- **Learned margin score with per-layer trainable β and γ (3.1).** g is frozen at 0: a run that also learned g (exp_g_0247, 1.169344) differed by 0.000016 bpb and by at most 0.016 in any layer's γ , so the gain was dropped. Trained γ 1.04–1.60, highest mid-depth; β 1.76–2.17 and nearly irrelevant to the score's shape. Both were still drifting at 16K.

Cell TV regularisation (outside Table A)

- **Penalty:** $\lambda \cdot \frac{1}{6} \sum_{\text{layers}} \frac{1}{P n 2^{n-1}} \sum_t \sum_{(c, c')} \|W_t[c] - W_t[c']\|^2$, squared L2 over the cell pairs (c, c') that differ in one bit; its gradient is added once per step, before gradient clipping. Training only: +6,144 FLOPs and +2,048 MACs per token per layer (amortised over 24,576 tokens); inference unchanged.
- **Orthogonal to the LUT math:** it acts on the tables alone, so it combines with every variant of all three generations, which doubles the number of runs. The current penalty (`lut_tv_penalty`) covers `LightMultiHeadLUT` tables only; Gen 1 and Gen 2 need it wired in.
- **One data point:** `exp_g_0249` = 3.1 + TV, $\lambda = 10$: **1.163698**, -0.0056 vs `exp_g_0248`, a mild regularisation benefit. The mechanism is mostly table shrinkage, not smoothing: table L2 norms fell to 0.22–0.35 $\times$ of `exp_g_0248`’s, while the scale-free ratio $\text{mean } \|W[c] - W[c']\|^2 / \text{mean } \|W[c]\|^2$ (2.0 for independent random cells) moved only from 1.74–1.95 to 1.34–1.57 across layers.

Where the code differs from the original brief

- 1.1’s input gradient is a surrogate (the derivative of 1.2’s blend); its weights update only the addressed row.
- The status report’s $u(\delta) = 0.5/(T + |\delta|)$ has a temperature; the code’s $0.5/(1 + |u|)$ has none (`l_projection.py:96`, `lutorch.cu:797`).
- Gen 2’s `hybrid_smooth` weight $\sigma(-2\rho_{j^*}/T_{sel})$ is a temperature-controlled soft-sig, not Gen 1’s $U(u)$.

Remaining work

- **Run 2.2, 2.3 and 2.4:** fork `exp_n_0121` verbatim, changing only `lut_forward_mode` / `lut_backward_topk`. No matching run exists (earlier `hybrid_smooth` runs are $n = 6$, tied, old eval; `exp_n_0188` is $n = 7$ with a gate on).
- **Run 3.2:** fork `exp_g_0248`, adding `lut_read_top_n: 2`, `lut_read_tau: 0.5`, `lut_read_tau_learnable: true`. No learned-score $n = 2$ run exists; `exp_g_0195` is $n = 2$ with the plain margin score, not this form.
- **Implement 1.1, 1.2:** wire Gen 1 into `CompressionMultiHeadLUT`, which accepts only `lut_impl` $\in \{\text{fast, light, bh4}\}$ (`compression_mhl.py:190`), as `MultiHeadLut` with `anchor_candidates` [128, 4, 48]. A hard-forward Gen-3 variant has not been designed.
- **Seeds:** zero replicates at this geometry; one extra seed of `exp_g_0248` (≈ 0.95 h) would give a real LUT seed spread. Reference: vanilla dense FFN 1.1651 / 1.1618 at 16K (two seeds), 1.1154 at 48K.